

UAVXGS — wiki/docs Report Index

Central registry for the simulation / tuning report lineage. Every report tracks its version, date, scope, and the simulator mechanics it reflects. When the simulation mechanics change, update the affected report(s) here and bump their version.

Rendering: `./md2pdf.sh <file> (pandoc → PDF, DejaVu Sans).`

Simulation & Calibration Series

Report	Version	Date	Sim mechanics covered	Status
PID_Simulation_v2.1	2.1	2026-08-04	Full sim construction: FW 3-axis coupled aero (quadratic damping, servo lag, mixing, stability/coupling), MR physics, AH/Nav outer loops, criteria, free-flight, known gaps. v2.1: is_roll_structurally_limited gate	Current — updated on every sim-mechanics change
FW_Control_Authority_Calibration_v2.1	2.1	2026-08-04	70%-servo-deflection → steady-state-rate authority anchor; PHYS_* .af storage; RATE_KP = 0.7/MAX_*_RATE; FC mixer paths per AF type; final tuned RATE_KP table	Current — anchor table updated when .af gains change
Cascade_Integrity_Checks_v1	1	2026-08-08	Attitude cascade safety cross-checks: outer IntLim ≤ max commanded rate (cascade windup), QP P-path demand ≤ max rate (loop separation), IntLim ≈ 20-30% of max rate as a <i>ceiling</i> ; GCS issues caveats (not clamps), sim test validates; caught the legacy IntLimit=10 migration bug	Current — caveats in parameter_window.compute_defaults checks in test_pid_sim.check_cascade

Tuning & Critique Series

Report	Date	Scope	Status
tuning_study.md	2026-08-04	v2.1 full pass: AH/Nav + attitude tuning across 12 FW + 5 MR generic frames; outer-loop bounds raise; AH_CRITERIA_FW 6s/8s; old-vs-revised tables; all 12 FW + 5 MR PASS	Current

Report	Date	Scope	Status
critique_report.md	2026-08-06	Critique of original/ frames — Phoenix FW retuned to PASS (v2.1); 9 MR frames legacy-baseline documented; v3.1: regenerated user/_Tuned fleet (misabeled FW frames fixed, get_fw_descriptor _Tuned-suffix fix); v4.0 (08-08): retired user/_Tuned → backup/_retired_tuned/ (fleet-wide IntLim=10 cascade-windup bomb, see Cascade_Integrity_Checks.md); FC ParamTable defaults canonical, mirrored into GCS PARAM_DEFAULTS	Current

GCS / Other Docs

Ecks_Airframe_Tuning.md, MC_TUNING_REPORT.md, GCS_Guide.md, GCS_Conversion.md, What_Is_Different_GCS.md.

Doc	Date	Scope	Status
Force_GPS_Init_Removal	2020-08-10	Force GPS Init removed (blocked main loop ~9.4 s → IWDG reset → USB loss); miscInitialiseGPS slot 9 reserved; GPS config bit default changed to ON in FC ParamTable + GCS defaults. Supp.1 (08-19): fc_build.py default board fixed → FLYINGRCF4WINGMINI (was wrong-target UAVXF4V3, USART1 GPS never fed); GPS packet decode + Kalman moved out of USART1 ISR into UpdateInertial (ISR = byte FSM + flag only). Supp.2 (08-20): GPS pass-through fully removed (FC + GCS, selector slot retained as unused); root cause of dead USART RX found — same-port RX pad never set to GPIO_Mode_AF in InitSerialPort (bulk analog init left it disconnected; USB targets unaffected); UAVXF4V3 GPS path (USART2/PA2-3, iNav UBX init) verified, UBX-only parser	Implemented — builds clean (FC fc_build.py OK, GCS py_compile OK)

Design Series

Doc	Version	Date	Scope	Status
ParamClass_Bounds_Design.md	0.0.1	2026-08-05	Class-based per-param bounds: FC tier-1 hard clamp, .af [LIMITS] tier-2 tuning range, GCS derived PARAM_LIMITS tier-3 fallback; full 128-param class mapping; migration plan	Draft — review before implementation

Soaring / Wind / ExcessLift Series

Doc	Date	Scope	Status
Soaring_Research_2026-08-06	2026-08-06	AnduPilot full auto-soar vs iNav motor-stop model; P0-P3 adoption plan for UAVX ArmQ	Research
ExcessLift_And_Wind_Implementation_Report	2026-10-06	Report on climb detector (control.c), Premierlani wind-triangle estimator (wind/wind.c), WPAItFail escape ladder (auto.c); rationale + emulator test plan	Implementation

Versioning Rules

1. **Sim mechanics change** (physics, mixing, criteria, authority anchor) → bump + update PID_Simulation_Construction_Report.md (and FW_Control_Authority_Calibration.md if authority is involved).
2. **Tuned gains change** → update FW_Control_Authority_Calibration.md RATE_KP table + tuning_study.md result tables.
3. Always keep the index table in sync so any report is resolvable from here.